

Definite integrals – 5

Integral identities and what they force upon the function

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Theoretical minimum.

Theorem 1 (moments determine a function). If f is continuous on $[a, b]$ and $\int_a^b x^n f(x) dx = 0$ for all $n \geq 0$, then $f \equiv 0$: by Weierstrass' theorem f is orthogonal to a uniformly dense family, hence to itself. For merely integrable f the same hypotheses give $f = 0$ almost everywhere. Finitely many vanishing moments say much less, but they do force sign changes: if $\int_a^b x^k f = 0$ for $k = 0, \dots, n-1$, then f changes sign at least n times.

Theorem 2 (an oscillating factor averages out). Let g be 1-periodic and integrable on $[0, 1]$ and let f be integrable on $[0, 1]$. Then

$$\lim_{n \rightarrow \infty} \int_0^1 f(x) g(nx) dx = \left(\int_0^1 f \right) \left(\int_0^1 g \right).$$

It suffices to check this for f an indicator of an interval and then to approximate. The case $g = \cos 2\pi x$ is the Riemann–Lebesgue lemma; for the rate of convergence see Definite integrals – 6.

Theorem 3 (Weyl's criterion). For a sequence (x_n) in $[0, 1]$ the following are equivalent: the averages $\frac{1}{n} \sum_{i \leq n} f(x_i)$ converge to $\int_0^1 f$ for every continuous f ; the same holds for every $f(x) = x^k$; the same holds for every $f(x) = e^{2\pi i k x}$, $k \neq 0$. Again the reason is that each of these families is dense, so a single uniform bound transfers the identity from the family to all of $C[0, 1]$. See Trig – 5 for the equidistribution of $\{n\alpha\}$.

Fact 1 (differentiate the identity). An identity of the form $\int_{a(x)}^{b(x)} f(t) dt = F(x)$, valid for all x , is a statement about f and should be differentiated: $f(b(x))b'(x) - f(a(x))a'(x) = F'(x)$. This both upgrades the regularity of f (the left side is automatically as smooth as F' allows) and turns the identity

into a functional equation for f itself; a scaling window $a(x) = 0$, $b(x) = \lambda x$ produces $\lambda f(\lambda x)$ and hence a multiplicative equation.

Fact 2 (integral sums with an error term). If $|f'| \leq M$ on $[0, 1]$, then $|\int_0^1 f - \frac{1}{n} \sum_{k=1}^n f(\frac{k}{n})| \leq \frac{M}{2n}$, and the midpoint sum improves this to $\frac{M_2}{24n^2}$ for $|f''| \leq M_2$. The general machinery is in Definite integrals – 5 and Limits – 6.

Fact 3 (a substitution that permutes the pieces). If φ takes the same value on several branches, the integral $\int f(\varphi(x))dx$ splits along them and the branches can be matched by a change of variable. The same idea reads an integral over a family of intervals as one integral over an orbit: the map $x \mapsto \frac{1}{1-x}$ has order three and $x^3 - 3x + 1$ is invariant under it.

1. (Putnam 1947) Let $f(x)$ be a differentiable function defined in the closed interval $(0, 1)$ and such that $|f'(x)| < M$, $0 < x < 1$. Prove that

$$\left| \int_0^1 f(x)dx - \frac{1}{n} \sum_{k=1}^n f\left(\frac{k}{n}\right) \right| \leq \frac{M}{n}.$$

2. (Putnam 1963)

- (i) Prove that if a function f is continuous on the closed interval $[0, \pi]$ and if

$$\int_0^\pi f(\theta) \cos \theta d\theta = \int_0^\pi f(\theta) \sin \theta d\theta = 0,$$

then there exist points α and β such that $0 < \alpha < \beta < \pi$ and $f(\alpha) = f(\beta) = 0$.

- (ii) Let R be any bounded convex open region in the Euclidean plane (that is, R is a connected open set contained in some circular disk, and the line segment joining any two points of R lies entirely in R). Prove with the help of part (i) that the centroid (center of gravity) of R bisects at least three distinct chords of the boundary of R .

3. (Putnam 1967) If f and g are continuous and periodic functions with period 1 on the real line, then

$$\lim_{n \rightarrow \infty} \int_0^1 f(x)g(nx) dx = \left(\int_0^1 f(x)dx \right) \left(\int_0^1 g(x)dx \right).$$

4. (Putnam 1972) Let $f(x)$ be an integrable function in $0 \leq x \leq 1$ and suppose

$$\int_0^1 f(x)dx = 0, \quad \int_0^1 xf(x)dx = 0, \quad \dots, \quad \int_0^1 x^{n-1}f(x)dx = 0 \quad \text{and} \quad \int_0^1 x^n f(x)dx = 1.$$

Show that $|f(x)| \geq 2^n(n+1)$ in a set of positive measure.

5. (Putnam 1993) Show that

$$\int_{-100}^{-10} \left(\frac{x^2 - x}{x^3 - 3x + 1} \right)^2 dx + \int_{1/101}^{1/11} \left(\frac{x^2 - x}{x^3 - 3x + 1} \right)^2 dx + \int_{101/100}^{11/10} \left(\frac{x^2 - x}{x^3 - 3x + 1} \right)^2 dx$$

is a rational number.